

Hematological response of *Tilapia (Oreochromis niloticus)* in Laundry wastewater

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ABSTRACT

The high concentration of detergent in aquatic ecosystem potentially affects the physiological condition of the fish by disrupting the respiration process and changing the concentration of blood components and blood chemistry. The purpose of this study was to determine the condition of the hematological parameters of tilapia (*Oreochromis niloticus*) exposed to wastewater from the laundry industry. Each treatment was stocked with five fish per aquarium (50 x 30 x 30 cm). This study used a completely randomized design (CRD) technique with six treatments and three replications: P0 (0%) as a control, P1 (1%), P2 (2%), P3 (3%), P4 (4%) and P5 (5%) with each treatment was exposed to a specific concentration of wastewater and residues. The results showed that the hemoglobin levels of treatments were decreasing, with the lowest mean of hemoglobin level found in the P2 (7.05 gr%), and the lowest concentration on the 30th day was 7.71 gr%. There were no significant effects of wastewater to the number of erythrocytes and leucocytes among treatments ($P>0.05$). While there were increasing levels of hematocrit with the largest mean of hematocrit levels found in the P4 treatment with a value of 24.11% and the largest mean on the 20th day of observation with a value of 23.51%. Wastewater from the laundry industry can affect the hematological condition of tilapia by decreasing the hemoglobin concentration and increasing the hematocrit levels that higher than the normal condition.

Keywords: detergent; hematology; laundry industry; tilapia; wastewater

INTRODUCTION

Detergent is a type of water-soluble surfactant, which is used to remove impurities from laundry in the household and laundry industries. The work process of the laundry industry is simple and straightforward: to dissolve detergents in water, because detergents have a better hardness than soap (Yuliani et al., 2015). Therefore, this industry will produce wastewater that contains detergent and is being discharged directly to the nearest aquatic environment.

One of the causes of water quality reduction is caused by detergent from the laundry industry wastewater. The negative influence of detergents on physical and chemical conditions of water can occur directly or indirectly. The effect of detergent and residues from wastewater on the environment is a decrease in the levels of dissolved oxygen in the water. Surfactant is one of the main ingredients of the detergent, where this surfactant can cause foams in the

water and forms a layer that inhibits the process of transferring oxygen from air to water (Dwinna Aliza & Sugito, 2014).

Fishes are commonly used as bioindicators of the effect of heavy metals contamination (Authman, 2015). Tilapia (*Oreochromis niloticus*) is one of the representative fishes that can be used as a bioindicator for assessment of water pollution, bioaccumulation, and to measure the risk of pollution to humans (Adel et al., 2016).

The accumulation of detergent from the laundry industry will cause a low supply of dissolved oxygen in the water. This condition will disrupt the breathing process in fish and causes the death of fish (Sabilu, 2010). This death can be occurred due to physiological deviations of blood components. Changes in blood components and blood chemistry, both qualitatively and quantitatively, can affect the condition of the fish, as blood has a vital function for transporting chemicals (e.g., hormones), waste products from body

metabolism, carbon dioxide and oxygen (Sabilu, 2010). The activities of cells, tissues, and organs require nutrients and oxygen. These materials can only be supplied if there is no deviation in blood circulation. Therefore, hematological conditions can be used as indicators to detect and determine the health status of fish (Sabilu, 2010).

There aren't much researches on the influence of wastewater and residues of the laundry industry on the hematological condition of tilapia, so further studies are necessary. A fundamental contribution of this research is expected to provide information to the public, the households, and the owners of the laundry enterprises about the impact of laundry wastewater and detergent residues on aquatic life and aquatic environment, mainly to fish.

MATERIALS AND METHODS

Research period and location of the experiment. Tilapia were obtained from the Undulako Fish Seed Center, Kolaka Regency, Southeast Sulawesi Province. The measurement of the hematological parameters of tilapia was carried out in the Integrated Laboratory of Sciences of the Sembilanbelas November Kolaka University. Filtered water from the local government-owned water utilities (PDAM) was used for the Tilapia rearing. Wastewater used in this research was originated from one of the laundry industries in Kolaka Regency.

Research Design. This research was conducted by using completely randomized design (CRD) method consisting of six treatments and three repetitions: P0 (0%), P1 (1%), P2 (2%), P3 (3%), P4 (4%) and P5 (5%) with each treatment was exposed to a specific concentration of wastewater and residues from laundry industry.

Four months old tilapia (5 ± 0.3 - 7 ± 0.2 g, 7 ± 0.4 - 9 ± 0.1 cm) were acclimatized in two larger aquarium (100x80x80 cm) before stocked in the experimental aquaria (50x30x30cm). After acclimatization, tilapia were weighed and measured so that the weights and length will be uniform. Then the tilapia were stocked to the 18 experimental aquaria with stocking density five fish/aquarium. Continuous aeration was

performed homogenously to maintain a stable oxygen concentration in each tank (Siburian et al., 2019). 5% of water volume was siphoned and exchanged daily to remove the uneaten feed and the fish feces.

Tilapia culture was conducted for 30 days. The food used is a commercial tilapia food with a protein content of 40% (Siburian et al. 2019). Tilapia was fed on a limited basis, twice a day, at 08.00 and 17.00 WITA. The feed is distributed evenly and is given up to 5% of the weight of tilapia per day.

Research Parameters. The parameters observed in this study include water quality, surfactant analysis of wastewater from the laundry industry, and hematological parameters. The water quality parameters consist of dissolved oxygen, temperature, and pH were measured every day during the study. The industrial detergent analysis consists of anionic and cationic surfactant test. The anionic surfactant test: the test tube was filled with 10 ml of methylene blue solution, 5 ml of chloroform was added, 1% of the detergent solution was added and stirred, then color changes were observed; and cationic surfactant test: 10 ml of 0.002% blue bromine phenol solution in Na acetate buffer pH 3.6-3.9 was added to the test tube, then 1% detergent was added, stirred until it was distributed uniformly, then the color of the solution was observed (Washil et al., 2009).

Meanwhile, the hematological parameters consist of hemoglobin concentration, total erythrocyte, total leukocyte, and hematocrit. The blood was sampled through the caudal vein near the tail between tilapia scales. Blood samples are slowly suctioned up to 2 ml each tail, then transferred to a 4 ml vacuum tube that has been moistened with anticoagulation. Blood samples were taken on the first day as a control, day 10, day 20, and day 30.

Hemoglobin concentration was measured by using the Sahli method. Total erythrocyte and total leukocyte were counted by using an improved Neubauer hemocytometer. The hematocrit measurement was conducted using microhematocrit tubes

that flow with blood and then centrifuged at a speed of 1500 rpm.

Statistical Analysis. The data obtained from the hematological observations of tilapia were analyzed by analysis of variance ANOVA and followed by Duncan test using IBM SPSS version 23 (SPSS, Ing, USA). Differences were considered as being significant at $p \leq 0.05$.

RESULTS AND DISCUSSION

Water Quality Analysis. Water quality data are secondary data that support primary data of the hematological parameters of tilapia during the study. The water quality observations in this study are consist of dissolved (DO), temperature and pH, in each treatment/concentration.

According to table 1, it is shown that water quality parameters of control P0 have not changed from beginning to the end of the experiment. There were some differences in water quality data between various concentration treatments. In general, DO values of treatments decreased during the 30 days of the study. In control P0, the DO value was quite stable from the beginning to the end of the study. In the treatment of P1, the DO value decreased, starting from day 10 to day 30. The pH value in P1 treatment has increased from day 10 to day 30. The other treatments of P2, P3, P4, and P5 showed a similar trend by decreasing the DO concentrations and increasing the pH. While the temperature of all treatments during the investigation tends to be stable.

Table 1. Water Quality During The Study

Treatment/ Concentration	Day 0			Day 10			Day 20			Day 30		
	Do (mg/L)	Temperature (°C)	pH	Do (mg/L)	Temperature (°C)	pH	Do (mg/L)	Temperature (°C)	pH	Do (mg/L)	Temperature (°C)	pH
0%	6,6	26,7	7,1	6,6	26,3	7,1	6,6	26,8	7,7	6,5	25,3	7,2
1%	6,7	26,6	7,2	6,4	26,5	7,4	5,2	26,5	7,6	4,3	23,4	7,8
2%	6,9	26,4	7,2	5,8	26,3	7,5	4	27,1	8	3,8	23,6	7,6
3%	6,8	26,6	7,1	6,1	26,5	7,5	4,2	26,5	7,6	3,6	24,8	8,3
4%	6,7	26,5	7,1	6,2	26,4	7,8	4,4	25,6	8,3	3,9	24,5	8,5
5%	6,6	26,2	7,1	5,8	25,7	7,4	4,8	26,2	8,2	3,7	23,1	8,3
Average	6,7	26,5	7,1	6,2	26,3	7,5	4,9	26,5	7,9	4,3	24,1	8,0

The results of this study indicate that wastewater treatment from the laundry industry reduces the water quality by lowering the DO value. The higher the concentration of treatment correlates to the lower DO value. Furthermore, the longer the duration of the study, the lower the DO value.

These water quality parameters are secondary data that become the supporting data of the primary data on the hematological parameter condition of tilapia. These data were obtained from DO, pH, and temperature measurements during the study from beginning to end. According to table 1, it is shown that there is a decrease in DO in higher concentration treatments. This is due to the effect of the wastewater from the laundry industry that causes a reduction in the transfer of DO, resulting in a decrease in the DO. The wastewater from the laundry industry contains detergents that the

increase in the detergents will accumulate the surfactants in the water bodies, cause problems in the sedimentation of water, and prevent the transfer of oxygen. Besides, the phosphate from detergents in the upper part of the river can also stimulate the growth of water weeds. The increase in the number of aquatic plants will cause an increase in phosphorus decomposition and inhibition of the exchange of oxygen in the water so that the levels of dissolved oxygen in the water will be very low (microaerophilic) (Patty et al., 2015). Water temperature during the study fluctuated during the study, except in control P0 at a concentration of 0%. An increase in water temperature also causes a reduction in DO levels in the water. The optimal temperature for tilapia growth is between 22-29 ° C (Zuhrawati, 2014). While the pH value of water at various treatments increase with the high detergent concentration, this can be explained because detergents have

bases chemical properties so that the pH always increases due to the nature of the detergent bases.

Surfactant analysis of wastewater from laundry industry. The anionic surfactant test showed a solid blue color in the chloroform layer, and the cationic surfactant test showed a natural blue color. This indicates that the detergent solution in the first press laundry industry used in this study contained concentrated surfactants. These results are consistent with the statement by (Smulders, 2002) that residues in the laundry detergent industry exist four main groups: builders around 70-80%, surfactants around 20-30%, additives around 2-8 % and bleaching about 1-2%.

Hemoglobin Concentration. Hemoglobin (Hb) is a red pigment that carries oxygen in red blood cells, which is a protein-rich in iron. The primary function of hemoglobin is the transport of O₂ and CO₂. Hemoglobin can also determine the level of immunity in fish because it is closely related to the oxygen-binding capacity of the blood (Nirmala et al., 2013). The results of the study show a decrease in hemoglobin levels in all treatments, except control P0, up to the 30th day of measurements. The treatments that have been exposed to the wastewater from the laundry industry have varied responses to the level of hemoglobin in each treatment.

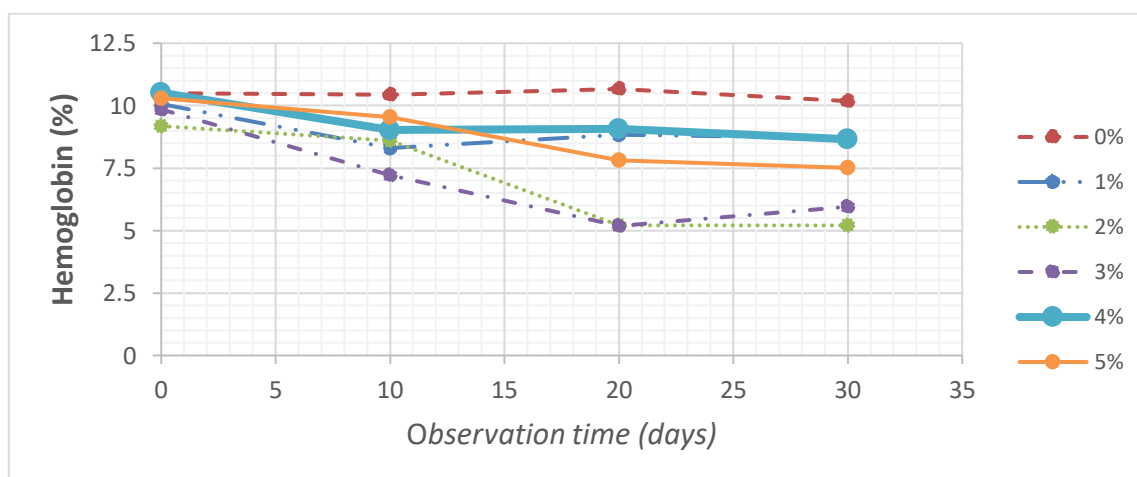


Figure 1. The average level of tilapia hemoglobin in treatment P0 (0%), P1 (1%), P2 (2%), P3 (3%), P4 (4%) and P5 (5%)

Figure 1 shows that the lowest hemoglobin levels were found in treatments P2, P3, P5, P1, and P4 with an average of hemoglobin levels, respectively 7.05 gr% 7.06 gr% 8.79 gr% 8.98 gr%, and 9.32 gr%. The control P0 had quite a constant hemoglobin level (10.45 gr%) since beginning to the end of the experiment. Statistical analysis showed that hemoglobin levels of P2 was not significantly different from P3 ($P > 0.05$), but was significantly different with three other concentrations (P1, P3, and P5) and control treatments ($P < 0.05$). The hemoglobin level of test fish in the concentration of P5 was not significantly different from P1 and P4 ($P > 0.05$) but was significantly different from the other two treatments (P2 and P3) and control treatments. The P0 control was significantly different from the other five treatments. This fact shows that the influence of wastewater in the

laundry industry can reduce hemoglobin levels in the blood of tilapia (Saparuddin & Arbain, 2019).

According to (Yasutake, 1977), normal hemoglobin levels in tilapia ranged from 10-11.1 g / dL. Hemoglobin levels below normal levels indicate that the hematopoiesis process in tilapia begins to be affected or hindered due to exposure of wastewater from the laundry industry.

The ability to bind to oxygen in the blood depends on the amount of hemoglobin present in the red blood cells. Low hemoglobin levels cause the metabolic rate to decrease, and the energy produced becomes depressed (Bastiawan et al., 1995). The main effect of lead is to affect heme synthesis, which causes damage to the blood. The measurement of hemoglobin can provide information on the impact of heavy metals on the physiological function of hemoglobin (Tunggul et al., 2005).

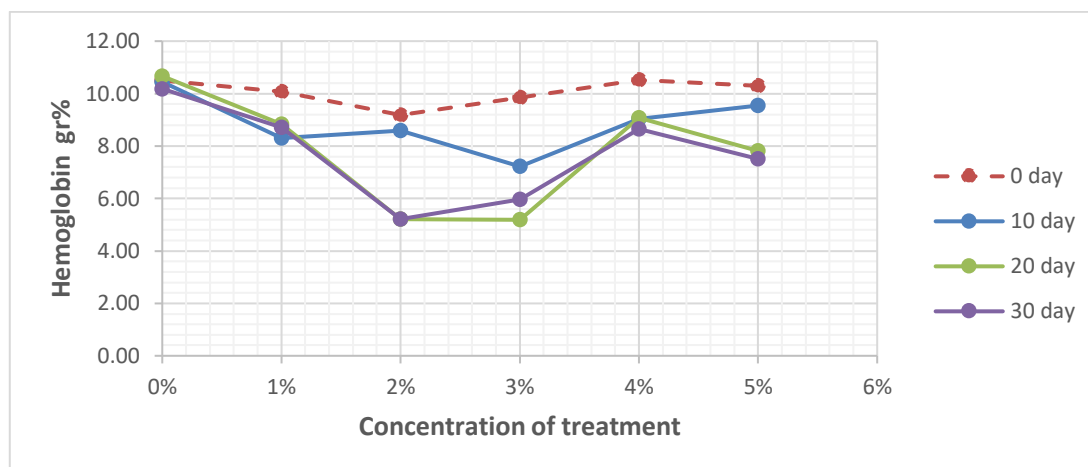


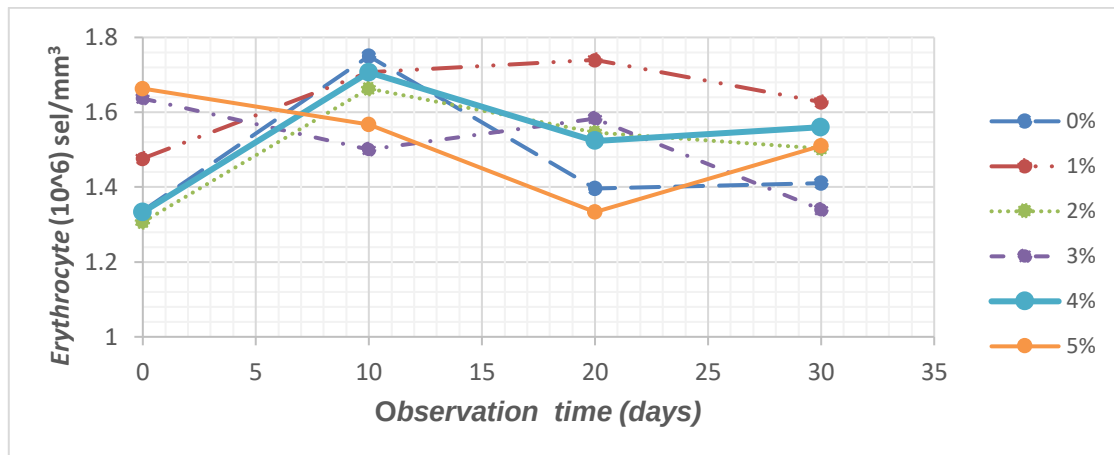
Figure 2. Average tilapia hemoglobin levels under observation day 0, day 10, day 20 and day 30

Figure 2 shows that the lowest hemoglobin level is in days 30, 20, 10, and 3, with the following percentage of 7.71 gr% 7.80 gr% 8.86 gr% and 10.07 gr%. Statistical analysis results showed that the hemoglobin level on day 30 did not differ with the hemoglobin level on day 20, but was different from the hemoglobin level on day ten and day 0 ($p < 0.05$). The hemoglobin level on day 20 was significantly different from the hemoglobin level on day ten and day 0. Similarly, the hemoglobin level on day 10 was significantly different from the apparent rate of hemoglobin on day 0 ($p < 0.05$). While the level of hemoglobin on day 0 varies in real with all observed days. This indicates that time exposure to the laundry waste detergent industry affects the level of hemoglobin tilapia.

Number of Erythrocytes. Erythrocytes are without core and have biconcave discs with a diameter of 8 μm thick edges of 2 μm , and the thickness of the middle section is reduced to 1 μm (Hartika et al., 2014). The main component of erythrocytes is hemoglobin protein that carries some oxygen (O_2) and carbon dioxide (CO_2) fraction.

Figure 3 shows that the amount of tilapia's erythrocytes is not subjected to significant changes in quantity with the presence of detergent from the laundry industry. The highest number of erythrocytes can be found in P1 (1.64×10^6 cells/ mm^3), followed by P4 (1.64×10^6 cells/ mm^3), P3 (1.53×10^6 cells/ mm^3), P5 (1.52×10^6 cells/ mm^3), P2 (1.51×10^6 cells/ mm^3), and the smallest can be found in control P0 (1.47×10^6 cells/ mm^3).

The results of the statistical analysis showed that the number of erythrocytes in tilapia among the treatment were not different ($p > 0.05$). The total amount of tilapia erythrocytes was not affected by the particular concentration of wastewater from the laundry industry. All treatments had the number of erythrocytes that were in the normal range. As reported by (Hartika et al., 2014), the average number of erythrocytes in tilapia ranges from 20,000 – 3 million cells/ mm^3 . The number of erythrocytes still within range numbers of health tilapia that indicates the hematopoiesis process is still happening in tilapia even though it has been exposed to the wastewater from the laundry industry.



Figure

Figure 3. The average number of erythrocyte tilapia in the treatment P0 (0%), P1 (1%), P2 (2%), P3 (3%), P4 (4%) and P5(5%)

Figure 4 shows that the number of erythrocytes at the lowest haemoglobin level is at days 0, 30, 20, and 10 with a consecutive number of 1.46×10^6 cells/mm³, 1.49×10^6 cells/mm³, 1.52×10^6 cells/mm³, and 1.65×10^6 cells/mm³. The results of the statistical analyses showed that the total number of tilapia erythrocytes that were linked with the wastewater from the laundry

industry on day 0 did not actually differ with the number of erythrocytes on the 30th and the 20th. But it's different from the 10th day. The number of erythrocytes on day 30 was not significantly different from day 20 and day 10. It shows that the wastewater from the laundry industry did not affect the number of tilapia erythrocytes.

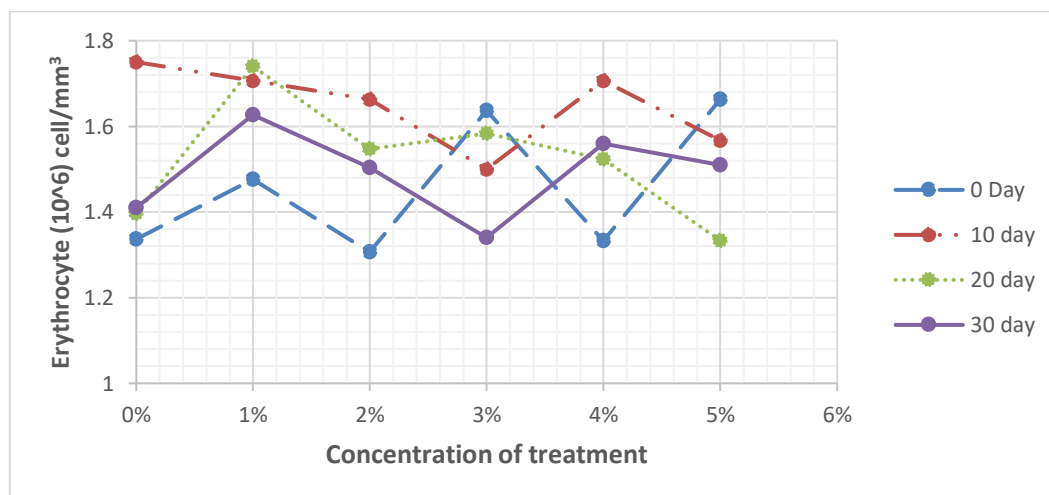


Figure 4. The average number of erythrocyte tilapia under observation Day 0, day 10, day 20 and day 30

Number of Leukocytes

Leukocytes in Teleost are a part of a non-specific body defense system. The decrease in the number of leukocytes is called leukopenia, while the increasing number of leukocytes is called leukocytosis (Uribe et al., 2011). Figure 5 shows that the number of tilapia leukocytes is aligned with the waste detergent laundry industry are not to undergo significant changes in term of

quantity. Where the smallest of leukocytes can be found in P2 (10.25×10^4 cells/mm³) followed P3 (11.00×10^4 cells/mm³), P4 (11.28×10^4 cells/mm³), P5 (11.39×10^4 cells/mm³), and the highest can be found in P0 (11.39×10^4 cells/mm³) and P1 (11.39×10^4 cells/mm³).

The results of the statistical analysis showed that the number of leukocytes in tilapia was not significantly different ($p > 0.05$) among

the treatments. All observations can be seen in the normal range of white blood cell count in regular fish generally ranges from 20,000 to 150,000 cells/mm³ (Sasongko, 2001). It shows

that the number of tilapia leukocytes was not affected by the exposure of wastewater from the laundry industry.

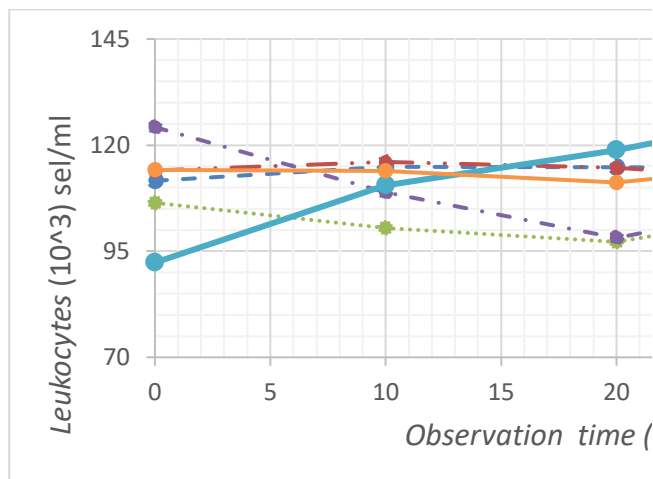


Figure 5. The average number of leukocytes of tilapia in the treatment P0 (0%), P1 (1%), P2 (2%), P3 (3%), P4 (4%) and P5 (5%)

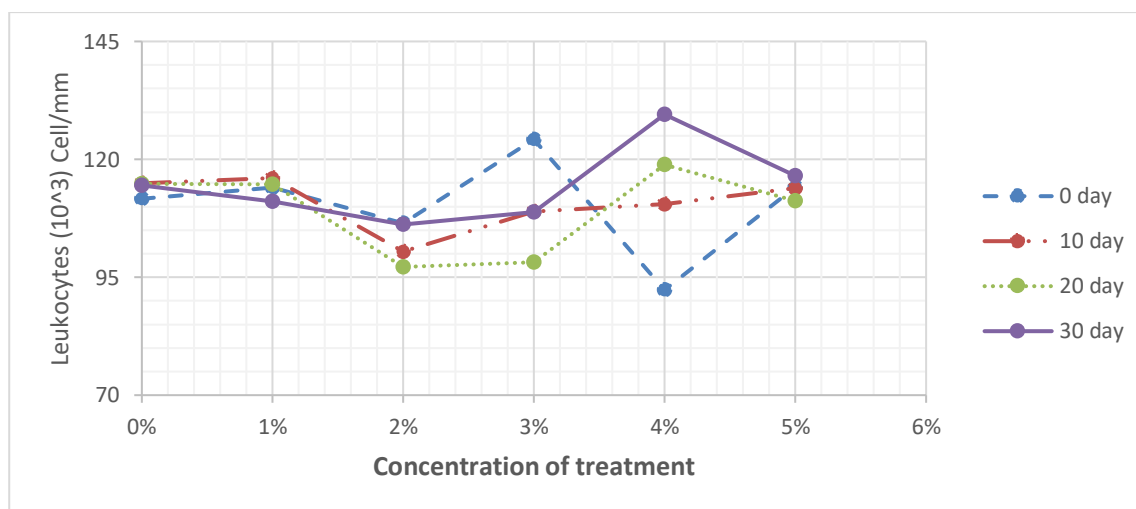


Figure 6. The average number of leukocytes tilapia under observation Day 0, day 10, day 20 and day 30

Figure 6 shows that the lowest number of leukocytes are found on day 20, day 0, day 10, and day 30 with a consecutive number of 10.91×10^4 cells/mm³, 11.04×10^4 cells/mm³, 11.07×10^4 cells/mm³, and 11.44×10^4 cells/mm³. Statistical analysis results show that the number of tilapia leukocytes that are lined with the waste laundry detergent industry from day 0 to day 30 has no real difference.

The number of leukocytes that are still in standard numbers indicates that the hematopoiesis process continues to occur in tilapia even though

it has been exposed to the waste detergent laundry industry. Several factors influence the number of leukocytes in fish, which consist of the type of species, age, and muscle activity. According to Dwinn Aliza & Sugito (2014), leukocytes will decrease if the fish is in a stressful condition, such as heat stress. Leukocytes will increase when fish become infected as a form of the immune response in the fight against microorganisms.

Hematocrit Level. Hematocrit level is a comparison between red blood cells and blood

plasma and affects the adjustment of red blood cells. This measurement is the percentage of erythrocytes in the whole blood after the blood sample is centrifuged (Fahmi Royan, Sri Rejeki, 2014).

Data on the results showed an increase in hematocrit levels across exposure to the

treatment of the laundry detergent industry up to the 30th-day measurement. The lowest hematocrit level can be found in control P0 (21.77%). The highest hematocrit level can be found in P4 (24.11%), followed by P5 23.64%, P2 (22.83%), P1 (21.77%).

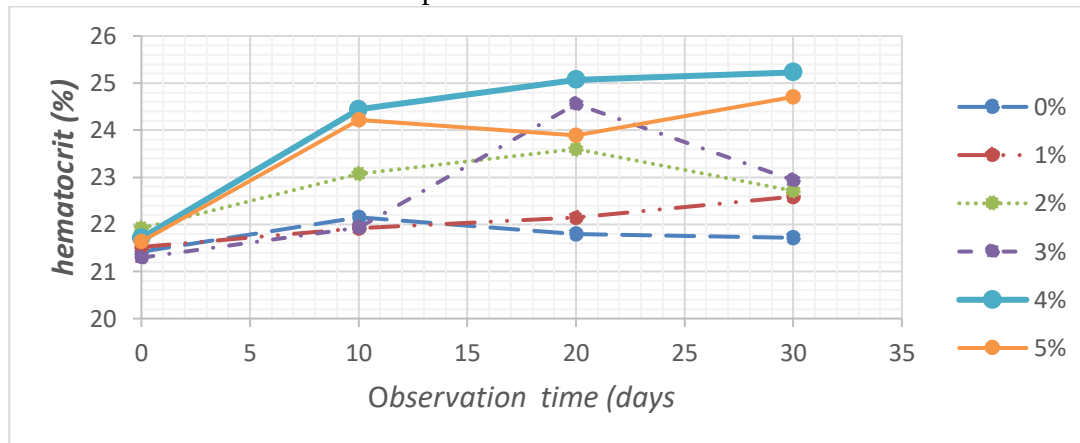


Figure 7. Average hematocrit value of tilapia in the treatment P0 (0%), P1 (1%), P2 (2%), P3 (3%), P4 (4%) and P5 (5%)

Figure 7 indicates that the hematocrit value at a level of P2 was not different from the concentration of P0, P1, P3, and P5 treatments but had a significant difference to P4 treatment ($P < 0.05$). The value of hematocrit in P0 was significantly different from the P4 ($p < 0.05$), but not to P3, P5, and P1 treatments. The P1 value was no different from the P3 and P5 treatment, but they were different from P4.

The results of statistical analyses show that the higher concentration of the wastewater of the laundry industry correlates to the higher increase of hematocrit level. According to (Fahmi Royan, Sri Rejeki, 2014), The normal hematocrit range

of tilapia is in between 21.00%-22.67%. The high value of hematocrit (above normal levels) indicates that the hematopoiesis process in tilapia began to be interrupted due to exposure from the wastewater of the laundry industry.

The calculation of the hematocrit value and hemoglobin level reflects the oxygen that carries the carrying power in the blood. A low concentration of hematocrit can cause damage or defects or osmoregulation proses, while a high value indicates an increased demand for oxygen or pressure (Dewi, 2012).

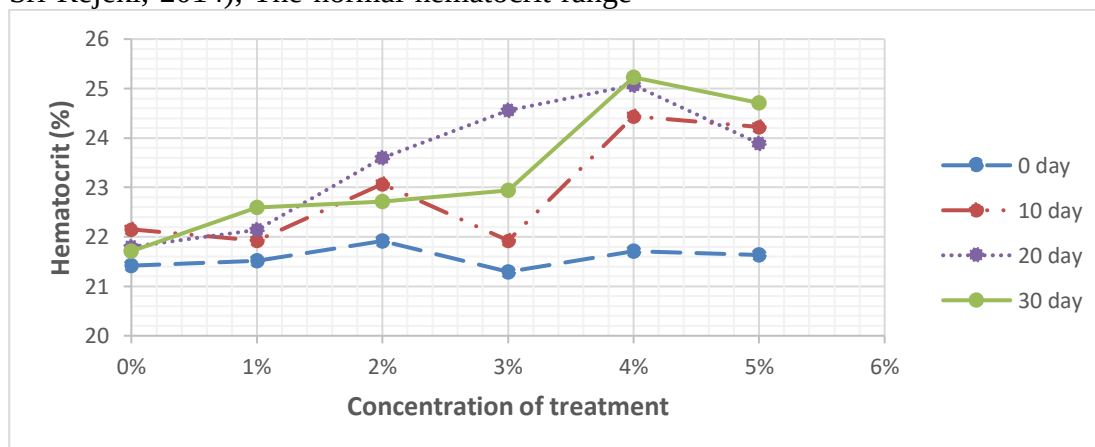


Figure 8. Average tilapia hematocrit value under observation day 0, day 10, day 20 and day 30

Figure 8 shows that the lowest

The hematocrit concentration on day 0, day 10, day 20, and day 30 are 21.58%, 22.96%, 23.31%, and 23.51% respectively. The results of statistical analyses showed that hematocrit tilapia levels are lined with the concentration of wastewater from the laundry industry since day 0 until day 30, and there were no significant differences among treatments ($p > 0.05$).

CONCLUSION

Wastewater from the laundry industry can affect the hematological condition of tilapia by decreasing the hemoglobin and increasing the hematocrit levels that higher than the normal condition.

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